

Solve geometry and statistics problems

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: A pie-chart sector is 126 degrees and represents 42 pupils. How many pupils are in the survey? Copy or complete the first step.

2. Create two different calculations using $\frac{3}{4}$, 0.6 and 25% that both give 30.

3. A quantity increases from 240 by 15%, then decreases by 20%. Find the final quantity.

4. Miro says: Miro starts calculating before deciding what the problem asks and which information matters. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: A pie-chart sector is 126 degrees and represents 42 pupils. How many pupils are in the survey? Copy or complete the first step.

Answer 120 pupils.

2. Create two different calculations using $\frac{3}{4}$, 0.6 and 25% that both give 30.

Answer Answers vary and must preserve the stated values and give 30.

3. A quantity increases from 240 by 15%, then decreases by 20%. Find the final quantity.

Answer 244.8.

4. Miro says: Miro starts calculating before deciding what the problem asks and which information matters. Is this correct? Explain.

Answer Represent the relationships first, label each subtotal and test the final answer against every condition.

Solve geometry and statistics problems

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. A pie-chart sector is 126 degrees and represents 42 pupils. How many pupils are in the survey?

6. Check this result using an inverse, estimate or known fact: A quantity increases from 240 by 15%, then decreases by 20%. Find the final quantity.

2. Create two different calculations using $\frac{3}{4}$, 0.6 and 25% that both give 30.

7. Prove or explain your reasoning: Three fifths of a number is 84. Find 35% of the number.

3. A quantity increases from 240 by 15%, then decreases by 20%. Find the final quantity.

8. Identify and correct this misconception: Miro starts calculating before deciding what the problem asks and which information matters.

4. Solve this independently and show every stage: A pie-chart sector is 126 degrees and represents 42 pupils. How many pupils are in the survey?

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Create two different calculations using $\frac{3}{4}$, 0.6 and 25% that both give 30.

10. Write and solve a similar question to: A quantity increases from 240 by 15%, then decreases by 20%. Find the final quantity.

Teacher answers and checking prompts

1. A pie-chart sector is 126 degrees and represents 42 pupils. How many pupils are in the survey?
Answer 120 pupils.
2. Create two different calculations using $\frac{3}{4}$, 0.6 and 25% that both give 30.
Answer Answers vary and must preserve the stated values and give 30.
3. A quantity increases from 240 by 15%, then decreases by 20%. Find the final quantity.
Answer 244.8.
4. Solve this independently and show every stage: A pie-chart sector is 126 degrees and represents 42 pupils. How many pupils are in the survey?
Answer 120 pupils.
5. Use a second method or representation for: Create two different calculations using $\frac{3}{4}$, 0.6 and 25% that both give 30.
Answer Expected result: Answers vary and must preserve the stated values and give 30. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: A quantity increases from 240 by 15%, then decreases by 20%. Find the final quantity.
Answer Expected result: 244.8. A valid check must be shown.
7. Prove or explain your reasoning: Three fifths of a number is 84. Find 35% of the number.
Answer 49. The explanation must show why the method works.
8. Identify and correct this misconception: Miro starts calculating before deciding what the problem asks and which information matters.
Answer Represent the relationships first, label each subtotal and test the final answer against every condition.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: A quantity increases from 240 by 15%, then decreases by 20%. Find the final quantity.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Solve geometry and statistics problems

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: Three fifths of a number is 84. Find 35% of the number.

6. Create a new example that tests the same idea as: A quantity increases from 240 by 15%, then decreases by 20%. Find the final quantity.

2. Prove the answer to this question, rather than only calculating it: A pie-chart sector is 126 degrees and represents 42 pupils. How many pupils are in the survey?

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Create two different calculations using $\frac{3}{4}$, 0.6 and 25% that both give 30.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 244.8.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro starts calculating before deciding what the problem asks and which information matters.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Three fifths of a number is 84. Find 35% of the number.
Answer 49. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: A pie-chart sector is 126 degrees and represents 42 pupils. How many pupils are in the survey?
Answer Expected result: 120 pupils. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Create two different calculations using $\frac{3}{4}$, 0.6 and 25% that both give 30.
Answer Expected result: Answers vary and must preserve the stated values and give 30. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 244.8.
Answer One valid reconstruction is: A quantity increases from 240 by 15%, then decreases by 20%. Find the final quantity. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro starts calculating before deciding what the problem asks and which information matters.
Answer Represent the relationships first, label each subtotal and test the final answer against every condition.
6. Create a new example that tests the same idea as: A quantity increases from 240 by 15%, then decreases by 20%. Find the final quantity.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.